

Lei Zhao

PhD of Computer Science

Hewlett Packard Labs, Hewlett Packard Enterprise
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Research and Expertise

Model & Optimization: Large Language Models (LLM), Convolutional Neural Networks (CNN), Quantization, Mixed-Precision, Efficient Inference, Model Deployment

Hardware & Architecture: Computer Architecture, AI Accelerators, Hardware-Software Co-design, Architecture Simulation, Performance and Energy Modeling, Memory Hierarchy, Emerging Memory Technology

Software & Framework: ML Compiler (MLIR), Dataflow and Mapping Optimization, Compute Graph Optimization, Runtime and Toolchain Integration

Skills

Programming C/C++, Python, CUDA, Chisel

Frameworks PyTorch, Hugging Face Transformers, MLIR, LLVM, gem5, Timeloop/Accelergy

Education

University of Pittsburgh

PH.D. OF COMPUTER SCIENCE

- Advised by Prof. Youtao Zhang

Pittsburgh, PA, U.S.

August 2014 - April 2022

Northwestern Polytechnical University

MASTER OF COMPUTER SCIENCE

- Advised by Prof. Danghui Wang

Xi'an, Shaanxi, China

August 2011 - April 2014

Northwestern Polytechnical University

BACHELOR OF SOFTWARE ENGINEERING

Xi'an, Shaanxi, China

August 2007 - July 2011

Work Experience

Hewlett Packard Enterprise, USA

HEWLETT PACKARD LABS

Postdoc Research Scientist

August 2023 - April 2026

AI and ML Engineer III

May 2026 - Present

- Architecture, algorithms, compiler and system design for ReRAM/CMOS hardware accelerators
- Full-stack ML accelerator for Transformers, responsibilities include:
 - Designed the architecture of an in-memory computing accelerator for Transformer models.
 - Optimized models and performed noise-aware fine-tuning for efficient deployment.
 - Developed an ISA and an LLVM/MLIR-based compiler targeting the accelerator.
 - Conducted performance and energy modeling via simulation, analyzing accelerator efficiency and scalability.

Meta, Inc., USA

REALITY LABS

Postdoc Research Scientist

January 2022 - August 2023

- Neural network model optimization and deployment on mobile hardware platforms.
- Cross-platform AR/VR demo system development (Windows, Linux, MacOS).
- ReRAM based in-memory computing accelerator design for graphics pipeline.
- Software simulation for graphics pipeline.
- Tools and skills: Pytorch, OpenGL(GLSL), OpenCV, Python, C/C++.

- Developed a privacy-preserving neural network training framework in C++ with CUDA acceleration.
- Maintaining previously developed privacy-preserving neural network training framework.
- Convert and optimize DNN model into a privacy-preserving friendly fashion.

PhD Projects

Machine Learning Accelerators

PROJECT OWNER / PRIMARY RESEARCHER

University of Pittsburgh

April 2017 - April 2022

- Design ASIC accelerator architectures based on existing or emerging memory technologies to improve performance and energy efficiency of machine learning computations with a focus on model security and user privacy.
- Key Achievements: Three conference papers in [ICCAD'17] [ICS'19] [DAC'20] and one U.S. patent.

Emerging Memory Design

PROJECT OWNER / PRIMARY RESEARCHER

University of Pittsburgh

September 2014 - April 2016

- Emerging memory system design for machine learning acceleration.
- Developed a cycle-accurate out-of-order architecture simulator based on MIPS ISA. Open sourced on github: <https://github.com/leizhaocs/Monichi>.
- Key Achievements: Four conference papers in [ICCD'15] [ISQED'17] [NVMSA'17] [ICCAD'17] and one journal paper in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD).

Publications

PHD THESIS

- **Lei Zhao**. (2022). [Secure Accelerator Design for Deep Neural Networks](#). University of Pittsburgh.

ACADEMIC PAPERS

- Archit Gajjar, Ruthwik Sunketa, **Lei Zhao**, Omar Eldash, Aishwarya Natarajan, Giacomo Pedretti, Aman Arora, Paolo Faraboschi, Jim Ignowski, and Luca Buonanno. (2026). [Azure-Lily: An FPGA Architecture with Analog IMC Engines for Efficient AI](#). ACM Transactions on Architecture and Code Optimization (TACO'26)
- **Lei Zhao**, Luca Buonanno, Archit Gajjar, John Moon, Aishwarya Natarajan, Sergey Serebryakov, Ron M. Roth, Xia Sheng, Youtao Zhang, Paolo Faraboschi, Jim Ignowski and Giacomo Pedretti. (2025). [NL-DPE: An Analog In-memory Non-Linear Dot Product Engine for Efficient CNN and LLM Inference](#). arXiv:2511.13950
- **Lei Zhao**, Aishwarya Natarajan, Luca Buonanno, Archit Gajjar, Ron Roth, Sergey Serebryakov, John Moon, Omar Eldash, Jim Ignowski and Giacomo Pedretti. (2025). [RACE-IT: A Reconfigurable Analog Computing Engine for In-Memory Transformer Acceleration](#). International Conference on Computer Design. (ICCD'25)
- London, Yanir, Luca Buonanno, Lei Zhao, Giacomo Pedretti, Bassem Tossoun, Stanley Cheung, Yuan Yuan, Yiwei Peng, Matej Hejda and Marco Fiorentino. (2025). [Optical content-addressable memories for tree-based machine learning](#). AI and Optical Data Sciences VI
- Archit Gajjar, **Lei Zhao**, Omar Eldash, Aishwarya Natarajan, Xia Sheng, Giacomo Pedretti, Paolo Faraboschi, Jim Ignowski, Aman Aro and Luca Buonanno. (2025). [Analog In-memory Computing Enhanced FPGA for High-Throughput and Energy-Efficient Acceleration](#). IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM'25)
- Archit Gajjar, **Lei Zhao**, Omar Eldash, Aishwarya Natarajan, Rand Jean, Xia Sheng, Giacomo Pedretti, Paolo Faraboschi, Jim Ignowski and Luca Buonanno. (2025). [Enhancing FPGAs with Analog In-Memory Computing Macros](#). Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (ISFPGA'25)
- **Lei Zhao**, Luca Buonanno, Aishwarya Natarajan, Jim Ignowski and Giacomo Pedretti. (2024). [Noise Aware Finetuning for Analog Non-Linear Dot Product Engine](#). ML with New Compute Paradigms (MLNCP) at NeurIPS

2024 (MLNCP workshop at NeurIPS 2024)

- Luca Buonanno, Giacomo Pedretti, **Lei Zhao**, Aishwarya Natarajan, Todd Richmond, John Moon, Rand Jean, Xia Sheng, Ron Roth and Jim Ignowski. (2024). **Memristive Quaternary Content-Addressable Memories for Implementing Boolean Functions**. IEEE International Symposium on Circuits and Systems (ISCAS'24)
- Giacomo Pedretti, John Moon, Pedro Bruel, Sergey Serebryakov, Ron M. Roth, Luca Buonanno, Archit Gajjar, **Lei Zhao**, Tobias Ziegler, Cong Xu, Martin Foltin, Paolo Faraboschi, Jim Ignowski and Catherine E. Graves. (2024). **X-TIME: accelerating large tree ensembles inference for tabular data with analog CAMs**. IEEE Journal of Exploratory Solid-State Computational Devices and Circuits (JXCDC'24)
- **Lei Zhao**, Luca Buonanno, Ron M. Roth, Sergey Serebryakov, Archit Gajjar, John Moon, Jim Ignowski and Giacomo Pedretti. (2023). **RACE-IT: A Reconfigurable Analog CAM-Crossbar Engine for In-Memory Transformer Acceleration**. (arXiv:2312.06532)
- **Lei Zhao**, Jae-sun Seo, H. Ekin Sumbul, Edith Beigne, and Dawei Wang. (2023). **ReARVR: A ReRAM-Based DNN Accelerator for Mobile Devices**. Design Automation Conference (Poster Session). (DAC'23)
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2022). **A DNN Protection Solution for PIM Accelerators With Model Compression**. IEEE Computer Society Annual Symposium on VLSI. (ISVLSI'22)
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2022). **SRA: A Secure ReRAM-Based DNN Accelerator**. Design Automation Conference. (DAC'22)
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2021). **Flipping Bits to Share Crossbars in ReRAM-Based DNN Accelerator**. International Conference on Computer Design. (ICCD'21)
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2020). **SCA: A Secure CNN Accelerator for both Training and Inference**. Design Automation Conference. (DAC'20)
- Zijie Yue, Shuai Ding, **Lei Zhao**, Youtao Zhang, Zehong Cao, M. Tanveer, Alireza Jolfaei, and Xi Zheng. (2020). **Privacy-preserving Time Series Medical Images Analysis Using a Hybrid Deep Learning Framework**. ACM Transactions on Internet Technology.
- **Lei Zhao**, Quan Deng, Youtao Zhang, and Jun Yang. (2019). **RFacc: A 3D ReRAM Associative Array based Random Forest Accelerator**. International Conference on Supercomputing. (ICS'19)
- Wen Wen, **Lei Zhao**, Youtao Zhang, and Jun Yang. (2019). **Exploiting In-memory Data Patterns for Performance Improvement on Crossbar Resistive Memory**. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2017). **AEP: An Error-bearing Neural Network Accelerator for Energy Efficiency and Model Protection**. International Conference On Computer Aided Design. (ICCAD'17)
- Wen Wen, **Lei Zhao**, Youtao Zhang, and Jun Yang. (2017). **Speeding Up Crossbar Resistive Memory by Exploiting In-memory Data Patterns**. International Conference On Computer Aided Design. (ICCAD'17)
- **Lei Zhao**, Youtao Zhang, and Jun Yang. (2017). **Mitigating Shift-Based Covert-Channel Attacks in Racetrack Last Level Caches**. Non-Volatile Memory Systems and Applications Symposium. (NVMSA'17)
- **Lei Zhao**, Lei Jiang, Youtao Zhang, Nong Xiao, and Jun Yang. (2017). **Constructing Fast and Energy Efficient 1TnR based ReRAM Crossbar Memory**. International Symposium on Quality Electronic Design. (ISQED'17)
- Xianwei Zhang, **Lei Zhao**, Youtao Zhang, and Jun Yang. (2015). **Exploit Common Source-Line to Construct Energy Efficient Domain Wall Memory based Caches**. International Conference on Computer Design. (ICCD'15)

Patents

- **Lei Zhao**, Luca Buonanno, and Giacomo Pedretti. (2024). **Analog content addressable memory for general computing**. U.S. Patent, 12,462,872.
- **Lei Zhao**, Giacomo Pedretti and Ron M Roth. (2024). **Gray encoding for analog content addressable memory in general computing**. U.S. Patent, 12,505,879.
- Luca Buonanno, **Lei Zhao**, and Giacomo Pedretti. (2024). **Analog computation of shift and add for dot product engines**. U.S. Patent, 18/651,338 (Allowed, to be issued).

- **Lei Zhao**, Luca Buonanno, and Giacomo Pedretti. (2024). **Programmable in-memory accelerator architecture for transformer models**. U.S. Patent, 18/772,940 (Allowed, to be issued).
- **Lei Zhao**, Luca Buonanno, Archit Gajjar, Giacomo Pedretti, and Jim Ignowski. (2024). **Recursive analog content addressable memory device**. U.S. Patent, 18/933,512 (Allowed, to be issued).
- Giacomo Pedretti, **Lei Zhao**, and Luca Buonanno. (2025). **Analog content addressable memory approximating activation functions**. U.S. Patent, 19/195,605 (Filed).
- Luca Buonanno, Archit Gajjar, **Lei Zhao**, Aishwarya Natarajan, Omar Eldash, Giacomo Pedretti, Paolo, Faraboschi, and Jim Ignowski. (2025). **Field programmable gate array with integrated analog in-memory computing dot product**. U.S. Patent, 19/067,179 (Filed).
- Luca Buonanno, Giacomo Pedretti, **Lei Zhao**, and Jim Ignowski. (2024). **Current-mode operation of analog content addressable memories**. U.S. Patent, 18/914,938 (Filed).
- Luca Buonanno, **Lei Zhao**, Archit Gajjar, Giacomo Pedretti, and Jim Ignowski. (2025). **An Inequality engine performing a recursive bitwise analysis**. U.S. Patent, 19/352,926 (Filed).
- Luca Buonanno, Yanir London, Matej Hejda, Giacomo Pedretti, **Lei Zhao**, Thomas Van Vaerenbergh, and Jim Ignowski. (2026). **Optical Inequality Solver**. U.S. Patent, 19/450,695 (Filed).
- Paul-Philipp Manea, Luca Buonanno, Aishwarya Natarajan, **Lei Zhao**, Todd Richmond, Giacomo Pedretti, and Jim Ignowski. (2026). **Analog Content Addressable Memory With Strong Arm Latched Memristor and Calibrated Sense Amplifier**. U.S. Patent, 19/643,005 (Filed).
- Luca Buonanno, **Lei Zhao**, Aishwarya Natarajan, Giacomo Pedretti, and Jim Ignowski. (2026). **Dot Product Engine with Integrated Analog-to-digital Converter**. U.S. Patent, 19/664,477 (Filed).
- Giacomo Pedretti, Fabian Bohm, **Lei Zhao**, Lucas Morais, Masoud Mohseni, and Thomas Van Vaerenbergh. (2026). **Programming and Control of Optimization solver Accelerators**. U.S. Patent, 64/061,369 (Filed).
- Youtao Zhang, **Lei Zhao**, and Jun Yang. (2022). **System and method of deploying an artificial neural network on a target device**. U.S. Patent, 11,531,877 B2.
- Shuai Ding, **Lei Zhao**, Shanlin Yang, Hao Wang and Zijie Yue. (2020). **The architecture, image processing method and process of an artificial intelligence chip for medical endoscope**. Chinese. Patent, CN108055454B.

Honors & Awards

2021	Best Paper Candidate , 39th International Conference on Computer Design (ICCD)	<i>Virtual Conference</i>
2017	Best Paper Candidate , 18th International Symposium on Quality Electronic Design (ISQED)	<i>Santa Clara, CA, USA</i>
2014	Outstanding Master Degree Thesis , Graduation Commencement of Northwestern Polytechnical University	<i>Xi'an, Shaanxi, China</i>